COMSATS University Islamabad
Department of Computer Science
Course Description Form (CDF)

Course Information

Course Code: **CSC301**	Course Title: **Design and Analysis of Algorithms**
Credit Hours: **3(3, 0)**	Lecture Hours/Week: **3**
Lab Hours/Week: **0**	Pre-Requisites: **CSC211-Data Structures**

Course Objectives:

Sr. #	Course Objective
CO-1	To develop a comprehensive understanding of fundamental algorithmic concepts such as time complexity, space complexity, and different classes of problems (e.g., P, NP, NP-complete).
CO-2	To develop the ability to design efficient algorithms using various strategies like divide and conquer, dynamic programming, greedy algorithms
CO-3	To acquire the skills to analyze the performance of algorithms in terms of time and space complexity and compare different algorithms for the same problem.
CO-4	To discuss rigorous correctness proofs for algorithms
CO-5	To develop the ability to critically evaluate and modify existing algorithms to improve their performance or adapt them to new problems.

Course Content:

This course is designed to provide knowledge of the principles and techniques used in the design and analysis of algorithms. Topics cover: Overview of Algorithm; Designing Techniques: Brute Force, Decrease and Conquer, Divide and Conquer, Transform and Conquer Technique, Dynamic Programming, Greedy Approach, Correctness of Algorithms, Analyzing Time Complexity of Iterative and Recursive Algorithms; and Computability.

Unit wise Major Topics:

Unit	Topic	No. of teaching hours
1.	Overview of Algorithm: Concepts, Properties, The Role of Algorithms in Computing, Algorithm Design & Analysis Process, Iterative Algorithm Design Issues, Top-Down Design, Design using Recursion.	4.5
2.	Brute Force (BF) Technique: Designing Algorithms for Sorting problem, Pattern Matching, Closest-Pair, and Convex-Hull Problem.	3
3.	Decrease and Conquer Technique: Designing Algorithm for Sorting Problem, Graph Traversal, Topological Sorting, Algorithms for Generating Combinatorial Objects, Decrease by-a-Constant Factor Algorithms, Variable Size Decrease Algorithms.	4.5
4.	Divide and Conquer Technique: Designing Algorithms for Sorting Problem, Closest-Pair, Convex-Hull Problem, and Matrix Multiplication Problem.	4.5
5.	Transform and Conquer Technique: Transformation to more convenient instance, Transformation to different representation, and Problem Reduction.	3
6.	Dynamic Programming (DP) Technique: Component & Properties; Designing Algorithms for Edit Distance; Longest Common Subsequence (LCS); Knapsack; and Matrix Chain Multiplication Problems.	6

7.	Greedy Approach: Algorithm for Data Compression, Coin Change Problem.	3
8.	Correctness of Algorithms: Pre-conditions, Post-conditions, Loop Invariant, Correctness of Iterative & Recursive Algorithms.	4.5
9.	Analysis of Algorithms: RAM Model, Asymptotic Notations, Worst, Best & Average Case Behavior of Algorithms; Complexity Classes; Solving Recurrence Relations: Substitution Method, Recurrence Tree Method, Master Method and Time & Space Tradeoffs.	7.5
10.	Computability: The Complexity Classes P & NP; and Introduction to NP Complete Problems.	4.5
Total Contact Hours		**45**

Mapping of CLOs and GAs

Sr.#	Unit #	Course Learning Outcomes	Blooms Taxonomy Learning Level	GA
CLO-1	1	Demonstrate an algorithmic approach to a given problem.	*Understanding*	2
CLO-2	2-7	Design new algorithms for different computational problems.	*Creating*	2,4
CLO-3	8	Prove correctness of an algorithm using loop invariant and induction.	*Applying*	2,3
CLO-4	9	Analyze best, average, and worst-case behaviors of an algorithm.	*Analyzing*	3
CLO-5	10	Explain the concept of various complexity classes with examples.	*Understanding*	2

CLO Assessment Mechanism

Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5
Quizzes	Quiz 1	Quiz 4	Quiz 2	Quiz 3	-
Assignments	-	Assignment 3 & 4	Assignment 1	Assignment 2	-
Mid Term Exam	Mid Term Exam	-	Mid Term Exam	Mid Term Exam	-
Final Term Exam	Final Term Exam				

Text and Reference Books

Textbook:

1. Introduction to the Design and Analysis of Algorithms, Levitin, A., Pearson, 2017.

Reference Book:

1. Introduction to Algorithms, Cormen, T. H., Leiserson, C.E., Rivest, R.L. & Stein, C., MIT Press, 2022.